

1	Course Name:	Information Technology Infrastructure														
	Course Code:	BMIS2113														
	Course Classification:	Elective (core)														
2	Synopsis:	The Information Technology Infrastructure course provides students with a comprehensive understanding of the fundamental components and principles that underpin modern IT systems and networks. In an increasingly digitized world, a robust and efficient IT infrastructure is critical to the success of organizations. This course equips students with the knowledge and skills necessary to design, implement, and maintain IT infrastructure solutions that support business operations and ensure data security.														
3	Name(s) of Academic Staff:	1	Refer to timetable													
		2														
		3														
4	Semester and Year offered:	Year Offered		Semester		Remarks: Refer to programme structure										
5	Credit Value:	3														
6	Pre-requisite/ co-requisite (if any):															
7	Course Learning Outcomes (CLO) 	CLO1	Describe the key components of IT infrastructure, including hardware, software, networks, and data centers. (C2, PLO1)													
		CLO2	Manage hardware and software resources, network software and configuration, and maintenance of IT infrastructure. (C3, PLO2)													
		CLO3	Design the IT infrastructure and IT server architecture options for an organisation. (C4, PLO2)													
8	Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods															
	Course Learning Outcomes	Programme Learning Outcomes (PLO)											Teaching Methods	Assessment Methods		
		PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11				
	CLO1	✓												L, T, P, NF2F	Test, Examination	
	CLO2		✓											L, T, P, NF2F	Examination	
	CLO3		✓											L, T, P, NF2F	Assignment	
	Mapping with MQF Cluster of Learning Outcomes	C1	C2													
	9	Transferable Skills (if applicable)														
(Skills learned in the course of study which can be useful and utilized in other settings)																
1		Cognitive skills														
2		Interpersonal skills														
	3															

Open-ended response (if any)

4

10

Distribution of Student Learning Time (SLT)

Note: This SLT calculation is designed for home grown programme only.

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**										Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)		
			Physical				Online/ Technology-mediated (Synchronous)						
			L	T	P	O	L	T	P	O			
1	Introduction to IT Infrastructure * The definition of IT infrastructure * The infrastructure model * Cloud computing and infrastructures	1,3	2	1	1	-						3	
2	Non Functional Attributes - Availability Concepts * Calculating availability * Sources of unavailability * Availability patterns - Performance * Perceived performance * Performance during infrastructure design * Performance of a running system * Performance patterns	1,3	2	1	1	-						4	
3	Datacenter building blocks - Datacenter availability - Datacenter performance - Datacenter security	1,3	2	1	1	-						3	
4	Networking - Network topologies - Networking building blocks - Network virtualization - Network availability - Network performance - Network security	2,3	6	3	3	-						4	
5	Storage - Storage building blocks - Storage availability - Storage performance - Storage security	2,3	4	2	2	-						4	
6	Compute Compute building blocks Compute availability Compute performance Compute security	2,3	4	2	2	-						4	
7	Operating systems - Popular operating systems - Operating System building blocks - Operating system availability - Operating system performance - Operating system security	2,3	4	2	2	-						4	

8	End user devices - End user device building blocks - Desktop virtualization - End user device availability - End user device performance - End user device security	2,3	2	1	1	-						3
9	IT Infrastructure Management - Infrastructure Deployment options - Automation - Documenting the infrastructure	2,3	2	1	1							4
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
SUB-TOTAL SLT:												89
Continous Assesement		%	Face-to-Face (F2F)				NF2F Independent Learning for Assessment (Asynchronous)					
			Physical		Online/ Technology- mediated (Synchronous)							
1	Assignment	40	1				12					
2	Test	10	1				5					
3												
4												
5												
SUB-TOTAL SLT:												19
Final Assesement		%	Face-to-Face (F2F)				NF2F Independent Learning for Assessment (Asynchronous)					
			Physical		Online/ Technology- mediated (Synchronous)							
1	Examination	50	2				10					
2												
3												
4												
5												
SUB-TOTAL SLT:												12
SLT for Assessment:												31
GRAND TOTAL SLT:												120
A	% SLT for F2F Physical Component: [Total F2F Physical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]											50.00
B	% SLT for Online & Independent Learning Component: [(Total F2F Online + Total Independent Learning) / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]											50.00
C	% SLT for All Practical Component: [% F2F Physical Practical + % F2F Online Practical]											11.67
C1	% SLT for F2F Physical Practical Component [Total F2F Physical Practical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]											11.67
C2	% SLT for F2F Online Practical Component [Total F2F Online Practical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]											

Please tick (v) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

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Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requirements imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	Nil
12	References (include required and further readings, and should be the most current)	1. Rajput, R. (2025). IT infrastructure unlocked: Drive digital transformation fast, revolutionize technology management: With best practices, tools & FAQs. Independently Published. 2. Cisco Networking Academy. 2026. Available from: https://www.netacad.com [26 May 2026]
13	Other additional information (if applicable)	Nil